
Instructions for Paper Submissions to AISTATS 2025: Supplementary Materials

1 Methods

Throughout the paper we compare four methods: Sequential Neural Posterior Estimation (SNPE) ?, Neural Likelihood Estimation (NLE) ?, NAIVE-ROMC, and R2OMC. In all cases, we use a simulation budget of up to 10,000 simulator executions. SNPE and NLE are chosen as representatives of the Neural Density Estimation (NDE) family. We opt for the sequential version of NPE (i.e., SNPE) due to its slightly better performance at a comparable computational cost. In contrast, we use the single-round version of NLE, as its sequential counterpart (SNLE) incurred significantly higher computational costs with a fractional improvement in the performance. Overall, in our experiments with a simulation budget of 10,000 runs, we observed no substantial difference between the sequential and non-sequential versions of the methods and since the focus of this paper is not on this comparison, we selected SNPE and NLE for convenience.

1.1 SNPE

Algorithm 1 Sequential Neural Posterior Estimation with atomic proposals ?

```
1: Set  $\tilde{p}_1(\theta) = p(\theta)$ 
2:  $c \leftarrow 0$ 
3: for  $r = 1$  to  $R$  do
4:   for  $j = 1$  to  $N$  do
5:      $c \leftarrow c + 1$ 
6:     Sample  $\theta_c \sim \tilde{p}_r(\theta)$ 
7:     Simulate  $x_c \sim p(x|\theta_c)$ 
8:   end for
9:    $V_r(\Theta) := \begin{cases} (\frac{c}{M})^{-1} & \text{if } \Theta = \{\theta_{b_1}, \theta_{b_2}, \dots, \theta_{b_M}\} \text{ and } 1 \leq b_1 < b_2 < \dots < b_M \leq c \\ 0 & \text{otherwise} \end{cases}$ 
10:   $\phi \leftarrow \arg \min_{\phi} \mathbb{E}_{\Theta \sim V_r(\Theta)} \left[ \sum_{\theta_j \in \Theta} -\log \tilde{q}_{x_j, \phi}(\theta_j) \right]$ 
11:  Set  $\tilde{p}_{r+1}(\theta) := q_F(x_o, \phi)(\theta)$ 
12: end for
13: return Samples from  $\hat{p}_R(\theta|x_o)$  using  $q_F(x, \phi)(\theta)$ 
```

Sequential Neural Posterior Estimation (SNPE) consists of three versions, named SNPE_A, SNPE_B, and SNPE_C in the SBI benchmark ?. In this work, we use **SNPE-C**, as introduced by ?.

SNPE_C mitigates the high variance in importance weights observed in SNPE_B by incorporating "atomic" proposals, which allow for flexible density estimators such as neural networks. The posterior is estimated by training a neural network $F(x, \phi)$ to approximate $p(\theta|x)$ using a density estimator $q_F(x, \phi)(\theta)$. The loss is computed based on an estimated proposal posterior, with the expectation taken over proposal sets. For further details, see ?.

In our experiments, we generally use $R = 2$ rounds, each with 5000 training samples: the first set of 5000 is drawn from the prior, while the second set is sampled from the approximate posterior learned after the first round. Occasionally, if the initial 5000 samples do not yield a reliable approximation and conditioning on $\mathbf{y}^o$ leads to instabilities, we use a single round ($R = 1$) of 10000 samples.

Algorithm 2 Sequential Neural Likelihood ?

```
1: Set  $\hat{p}_0(\theta|x_o) = p(\theta)$  and  $D = \{\}$ 
2: for  $r = 1$  to  $R$  do
3:   for  $n = 1$  to  $N$  do
4:     Sample  $\theta_n \sim \hat{p}_{r-1}(\theta|x_o)$  using MCMC
5:     Simulate  $x_n \sim p(x|\theta_n)$ 
6:     Add  $(\theta_n, x_n)$  to  $D$ 
7:   end for
8:   (Re-)train  $q_\psi(x|\theta)$  on  $D$ 
9:   Set  $\hat{p}_r(\theta|x_o) \propto q_\psi(x_o|\theta)p(\theta)$ 
10: end for
11: return Samples from  $\hat{p}(\theta|x_o) \propto q_\psi(x_o|\theta)p(\theta)$  using MCMC;  $q_\psi(x|\theta)$ 
```

1.2 NLE

NLE algorithm is presented in Algorithm ?? . For more in-depth information, we direct the reader to Section A.3 of the Appendix in ? . Below, we offer a brief summary.

NLE approximates the likelihood $p(x_o|\theta)$ using neural density estimation. The process involves generating a dataset $D = \{(\theta_n, x_n)\}_{n=1}^N$ by sampling $\theta_n \sim p(\theta)$ and simulating $x_n \sim p(x|\theta_n)$. A conditional neural density estimator $q_\psi(x|\theta)$ is then trained to model the likelihood.

Once trained, samples from the posterior $\hat{p}(\theta|x_o) \propto q_\psi(x_o|\theta)p(\theta)$ are obtained using MCMC. For density estimation, a Masked Autoregressive Flow (MAF) with five flow transforms and 50 hidden units per block is employed. For MCMC, 250 warm-up steps and posterior samples are obtained with thinning.

2 Introductory example

We here provide analytical derivations and further evidence for the introductory example of Figure ?? .

The simulator samples from $\frac{1}{2}(\mathcal{N}(\boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(-\boldsymbol{\theta}, \mathbf{I}))$ and the prior is $p(\boldsymbol{\theta}) = \mathcal{U}(\boldsymbol{\theta}; -\mathbf{15}, \mathbf{15})$, where $\boldsymbol{\theta} \in \mathbf{R}^D$.

Simple Case:

Here, $D = 5$ and there is one observation $\mathbf{y}^o = (5, \dots, 5)$. The ground truth posterior is:

$$p(\boldsymbol{\theta}|\mathbf{y}^o) \propto p(\boldsymbol{\theta})p(\mathbf{y}^o|\boldsymbol{\theta}) \tag{1}$$

$$\propto \mathcal{U}(-\mathbf{15}, \mathbf{15}) \left(\frac{1}{2} \mathcal{N}(\mathbf{y}^o; \boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(\mathbf{y}^o; -\boldsymbol{\theta}, \mathbf{I}) \right) \tag{2}$$

$$\propto \begin{cases} \mathcal{N}(\mathbf{y}^o; \boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(\mathbf{y}^o; -\boldsymbol{\theta}, \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^2 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

$$\propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}^o, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{y}^o, \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^2 \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

$$\propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^2 \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

High-dimensional Case: We here change the dimensionality of the parameter space to $D = 20$ and the observation becomes $\mathbf{y}^o = (5, \dots, 5)$. Using the same derivations as in the simple case, the ground truth posterior is:

$$p(\boldsymbol{\theta}|\mathbf{y}^o) \propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^{20} \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

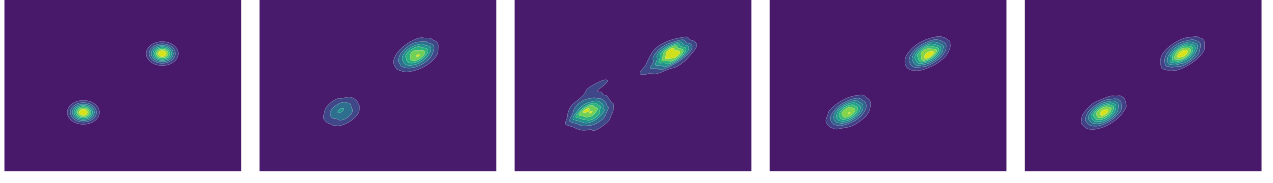


Figure 1: Caption

$p(\theta|\mathbf{y}^o) \propto \mathcal{N}(\theta; \mathbf{5}, \mathbf{I}) + \mathcal{N}(\theta; -\mathbf{5}, \mathbf{I})$, where $\theta \in \mathbb{R}^{20}$.

Distractors Case: Here the simulator samples from $\mathbf{y} \sim (\mathbf{y}^{(1)}, \mathbf{y}^{(2)})$

Algorithm 3 OMC. Requires $p(\theta)$, $g(\theta, \mathbf{u})$, S, ϵ

```

1: procedure OMC
2:   for  $i \leftarrow 1$  to  $S$  do
3:      $\mathbf{u}_i \sim p(\mathbf{u})$  ▷ Draw nuisance variables
4:      $d_i(\theta) = d(g(\theta, \mathbf{u}_i), \mathbf{y}^o)$  ▷ Define distance function
5:      $\theta_i^* = \operatorname{argmin}_{\theta} d_i(\theta)$  ▷ Solve optimization problem
6:     if  $d_i(\theta_i^*) > \epsilon$  then
7:       Go to 2 ▷ Filter solution
8:     end if
9:     Compute the Jacobian  $\mathbf{J}_i$  with columns  $\partial g(\theta_i^*, \mathbf{u}_i)/\partial \theta$  ▷ Compute Jacobian
10:    Compute  $w_i = p(\theta_i^*) * \det(\mathbf{J}_i^T \mathbf{J}_i)^{-1/2}$  ▷ Compute weight
11:    Accept  $\theta_i^*$  as posterior sample with weight  $w_i$  ▷ Accept sample
12:  end for
13:  Compute  $\mathbb{E}_{p_\epsilon}[h(\theta)] = \sum_{i=1}^S w_i h(\theta_i^*) / \sum_{i=1}^S w_i$  ▷ Estimate an expectation
14: end procedure

```

Algorithm 4 ROMC. Requires $p(\theta)$, $g(\theta, \mathbf{u})$, S, M, ϵ

```

1: procedure ROMC
2:   for  $i \leftarrow 1$  to  $S$  do
3:      $\mathbf{u}_i \sim p(\mathbf{u})$  ▷ Draw nuisance variables
4:      $d_i(\theta) = d(g(\theta, \mathbf{u}_i), \mathbf{y}^o)$  ▷ Define distance function
5:      $\theta_i^* = \operatorname{argmin}_{\theta} d_i(\theta)$  ▷ Solve optimization problem
6:     if  $d_i(\theta_i^*) > \epsilon$  then
7:       Go to 2 ▷ Filter solution
8:     end if
9:     Define  $q_i$  based on Algorithm ?? and compute  $V_i$  ▷ Estimate proposal area
10:    for  $j \leftarrow 1$  to  $M$  do
11:       $\theta_{ij} \sim q_i$  ▷ Sample
12:      Compute  $w_{ij} = p(\theta_{ij}) * V_i * \mathbb{1}_{C_\epsilon}(\theta_{ij})$  ▷ Compute weight
13:    end for
14:  end for
15:  Compute  $\mathbb{E}_{p_\epsilon}[h(\theta)] = \sum_{i=1}^S \sum_{j=1}^M w_{ij} h(\theta_{ij}) / \sum_{i=1}^S \sum_{j=1}^M w_{ij}$  ▷ Estimate an expectation
16: end procedure

```

3 Additional information on the experiments

All experiments were conducted on a single Dell XPS PC equipped with an Intel® Core™ i7-10750H CPU @ 2.60GHz with 12 cores.

Compared Methods. The methods that are used in the experiments are the following:

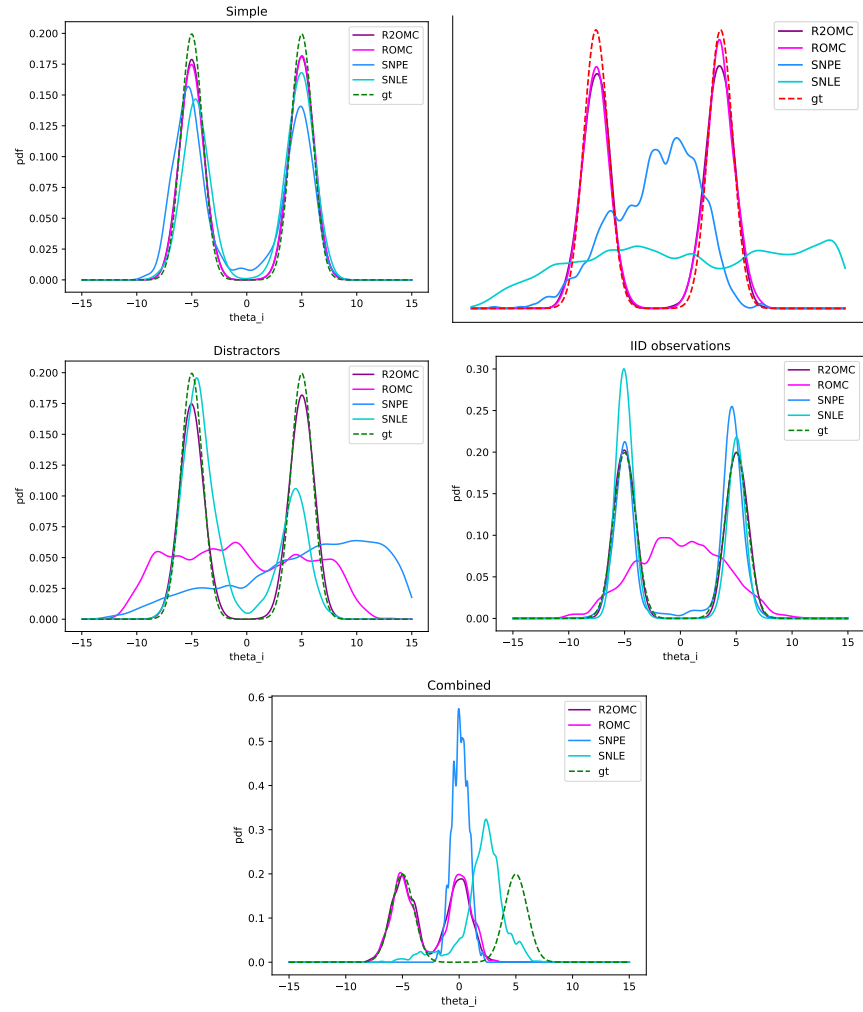


Figure 2: Caption

Algorithm 5 Hyperbox computation

Requires: θ^* optimal point, $d(\theta)$ distance function, $\mathbf{J}$ jacobian at θ^*

Parameters: η step size, R number of refinements, S number of steps, ϵ distance threshold

```
1: Compute eigenvectors  $\mathbf{v}_d$  of  $\mathbf{J}^T \mathbf{J}$  ( $d = 1, \dots, D$ )
2: for  $d \leftarrow 1$  to  $D$  do
3:    $\tilde{\theta} \leftarrow \theta^*$  ▷ Initialize the end point
4:    $r \leftarrow 0$  ▷ Refinements counter
5:   repeat
6:      $s \leftarrow 0$  ▷ Step counter
7:     repeat
8:        $s \leftarrow s + 1$  ▷ Increase step counter
9:        $\tilde{\theta} \leftarrow \tilde{\theta} + \eta \mathbf{v}_d$  ▷ Move a step  $\eta$ 
10:    until  $d(\tilde{\theta}) > \epsilon$  or  $s = S$  ▷ Stop if distance over  $\epsilon$  or reached maximum steps
11:     $\tilde{\theta} \leftarrow \tilde{\theta} - \eta \mathbf{v}_m$  ▷ Move a step back
12:     $\eta \leftarrow \eta/2$  ▷ Double the resolution
13:     $r \leftarrow r + 1$  ▷ Increase refinement counter
14:  until  $r = R$  ▷ Stop if maximum refinements
15:  Set  $\tilde{\theta}$  as end point.
16:  Repeat steps 3-15 for  $\mathbf{v}_d = -\mathbf{v}_d$  and set  $\tilde{\theta}$  as the negative end point along  $\mathbf{v}_d$ 
17: end for
18: Define uniform distribution  $q$  on top of the hyperbox
19: return  $q$ 
```

Algorithm 6 CoreR2OMC. Requires: $p(\theta), g(\theta, \mathbf{u}), \mathbf{y}^o$,

```
1: procedure CORER2OMC( $S, \epsilon_1, \epsilon_2, \eta, R, L$ )
2:   for  $i \leftarrow 1$  to  $S$  do
3:      $\mathbf{u}_i \sim p(\mathbf{u})$ 
4:      $\theta_i^* = \text{argmin}_{\theta} d(g(\theta, \mathbf{u}_i) \odot, \mathbf{y}^o \odot)$ 
5:     if  $d_i^*(\theta) > \epsilon_1$  then go to line 2
6:      $q_i$  around  $\theta_i^*$  with Alg. ?? using  $\eta, R, L, \epsilon_2$ 
7:   end for
8:   return  $\{q_i\}_i^S$ 
9: end procedure
```

- **R2OMC:** The proposed method. The exact implementation is provided at
- **OMC:** The original OMC method. The exact implementation is provided at
- **NPE:** The Neural Posterior Estimation method. We use the SNPE-C implementation provided (here) with `density_estimation='nsf'` and `hidden_units=50`.

3.1 Gaussian benchmark

We present analytical derivations for the ground truth posteriors of the Gaussian Benchmark. These posteriors are either plotted directly for visual inspection or used to generate ground truth samples, which are then compared with the samples produced by the methods under evaluation.

3.1.1 Multiple observations

Ground Truth. The simulator is $S_{\text{shift}} : \mathbf{y} \sim \frac{1}{2} (\mathcal{N}(\theta + \mathbf{5}, \mathbf{I}) + \mathcal{N}(\theta, \mathbf{I}))$, the prior is $p(\theta) = \mathcal{U}(-15, 15)$ and the set of N iid observations $\{\mathbf{y}_n^o\}_n^N$. The ground truth posterior is:

$$p(\boldsymbol{\theta}|\{\mathbf{y}_n^o\}) \propto p(\boldsymbol{\theta})p(\{\mathbf{y}_n^o\}|\boldsymbol{\theta}) \quad (7)$$

$$\propto p(\boldsymbol{\theta}) \prod_{n=1}^N p(\mathbf{y}_n^o|\boldsymbol{\theta}) \quad (8)$$

$$\propto \mathcal{U}(-\mathbf{15}, \mathbf{15}) \prod_{n=1}^N \frac{1}{2} (\mathcal{N}(\mathbf{y}_n^o; \boldsymbol{\theta} + \mathbf{5}, \mathbf{I}) + \mathcal{N}(\mathbf{y}_n^o; \boldsymbol{\theta}, \mathbf{I})) \quad (9)$$

$$\propto \mathcal{U}(-\mathbf{15}, \mathbf{15}) \prod_{n=1}^N \frac{1}{2} (\mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o - \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o, \mathbf{I})) \quad (10)$$

$$\propto \begin{cases} \prod_{n=1}^N (\mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o - \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o, \mathbf{I})) & \text{if } \boldsymbol{\theta} \in [\mathbf{15}, \mathbf{15}] \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

In ??, the factorization is due to the N independently and identically distributed (iid) data points.

In ??, we apply the property: $\mathcal{N}(\mathbf{y}; \boldsymbol{\theta} + \boldsymbol{\alpha}, \Sigma) = \mathcal{N}(\boldsymbol{\theta}; \mathbf{y} - \boldsymbol{\alpha}, \Sigma)$. In our simulations, the data are sufficiently far from the boundaries so that we may ignore the boundary effects in the truncated distribution above.

Case 1: Given N similar observations $\mathbf{y}_n^o \approx (0, \dots, 0)$ for $n \in \{1, \dots, N\}$, the posterior, for $\boldsymbol{\theta}$ sufficiently far from the boundary, is approximately:

$$p(\boldsymbol{\theta}|\mathbf{y}_n^o) \propto \prod_n^N (\mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o - \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o, \mathbf{I})) \quad (12)$$

$$\propto (\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I}))^N \quad (13)$$

$$\propto \sum_{n=0}^N \binom{N}{n} (\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}))^n (\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I}))^{N-n} \quad (14)$$

$$\propto (\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}))^N + (\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I}))^N \quad (15)$$

$$\propto \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \frac{1}{N}\mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \frac{1}{N}\mathbf{I}) \quad (16)$$

In ??, for $n \in \{1, \dots, N-1\}$ the terms involve products between $\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I})$ and $\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})$ so they have a much smaller magnitude compared to $(\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}))^N$ ($n = N$) and $\mathcal{N}((\boldsymbol{\theta}; \mathbf{0}, \mathbf{I}))^N$ ($n = 0$). This allows us to approximate the expression by retaining only these two dominant terms. In ??, we use the property that raising a Gaussian distribution to the power N results in a Gaussian with the same mean and a covariance scaled by $\frac{1}{N}$. In Figure ?? we show two dimensions, $\boldsymbol{\theta}_i$ and $\boldsymbol{\theta}_j$, of the ground truth posterior for $N = \{1, 4, 10, 20\}$ observations.

Case 2: Given an even number of N observations that are split in two categories, half around $(0, \dots, 0)$, i.e., $\mathbf{y}_n^o \approx (0, \dots, 0)$ for $n \in \{1, \dots, N/2\}$ and the other half around $(5, \dots, 5)$, i.e., $\mathbf{y}_n^o \approx (5, \dots, 5)$ for $n \in$

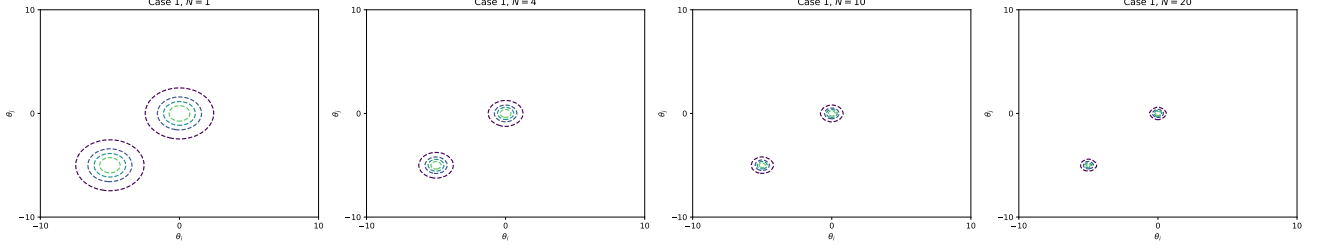


Figure 3: Multiple Observations - Case 1: Ground truth posterior for a varying number of observations $N \in \{1, 4, 10, 20\}$ from left to right. We observe that the Gaussian modes shrink with covariance $\frac{1}{N}\Sigma$ as the number of observations increases.

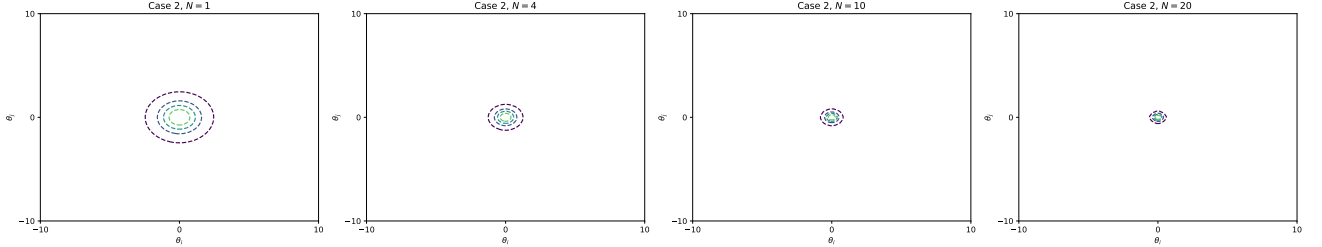


Figure 4: Multiple Observations - Case 2: Ground truth posterior for a varying number of observations $N \in \{1, 4, 10, 20\}$ from left to right. We observe that the Gaussian modes shrink with covariance $\frac{1}{N}\Sigma$ as the number of observations increases.

$\{N/2 + 1, \dots, N\}$, the posterior, for values of θ sufficiently far from the boundary, can be written as:

$$p(\theta|\{\mathbf{y}_k^o\}) \propto \prod_n^N (\mathcal{N}(\theta; \mathbf{y}_n^o - \mathbf{5}, \mathbf{I}) + \mathcal{N}(\theta; \mathbf{y}_n^o, \mathbf{I})) \quad (17)$$

$$\propto \prod_{n=1}^{N/2} (\mathcal{N}(\theta; -\mathbf{5}, \mathbf{I}) + \mathcal{N}(\theta; \mathbf{0}, \mathbf{I})) \prod_{n=N/2+1}^N (\mathcal{N}(\theta; \mathbf{0}, \mathbf{I}) + \mathcal{N}(\theta; \mathbf{5}, \mathbf{I})) \quad (18)$$

$$\propto \prod_{n=1}^{N/2} (\mathcal{N}(\theta; -\mathbf{5}, \mathbf{I}) + \mathcal{N}(\theta; \mathbf{0}, \mathbf{I})) \prod_{n=1}^{N/2} (\mathcal{N}(\theta; \mathbf{0}, \mathbf{I}) + \mathcal{N}(\theta; \mathbf{5}, \mathbf{I})) \quad (19)$$

$$\propto \left((\mathcal{N}(\theta; -\mathbf{5}, \mathbf{I}))^{N/2} + (\mathcal{N}(\theta; \mathbf{0}, \mathbf{I}))^{N/2} \right) \left((\mathcal{N}(\theta; \mathbf{0}, \mathbf{I}))^{N/2} + (\mathcal{N}(\theta; \mathbf{5}, \mathbf{I}))^{N/2} \right) \quad (20)$$

$$\propto \mathcal{N}(\theta; \mathbf{0}, \mathbf{I})^N \quad (21)$$

$$\propto \mathcal{N}(\theta; \mathbf{0}, \frac{1}{N}\mathbf{I}) \quad (22)$$

In ??, we keep only the dominant terms, as in Case 1. In ??, we use that terms that involve combinations of $\mathcal{N}(\theta; \mathbf{0}, \mathbf{I})^{N/2}$, $\mathcal{N}(\theta; -\mathbf{5}, \mathbf{I})^{N/2}$ and $\mathcal{N}(\theta; \mathbf{5}, \mathbf{I})^{N/2}$ have significantly smaller magnitudes compared to the dominant ones; $\mathcal{N}(\theta; \mathbf{0}, \mathbf{I})^{N/2}\mathcal{N}(\theta; \mathbf{0}, \mathbf{I})^{N/2}$. In ??, we use the property that raising a Gaussian distribution to the power N results in a Gaussian with the same mean and a covariance scaled by $\frac{1}{N}$. In Figure ?? we show two dimensions, θ_i and θ_j , of the ground truth posterior for $N = \{1, 4, 10, 20\}$ observations.

Results. Figure ?? demonstrates that R2OMC samples align closely with the ground truth posterior, correctly capturing the two modes around $(0, \dots, 0)$ and $(-5, \dots, 5)$. Due to the approximation error introduced by ϵ , the R2OMC samples exhibit slightly higher variance compared to the ground truth. In contrast, both ROMC and NPE struggle with multiple observations, resulting in their samples being distributed along the 45° line. A similar pattern is observed in Figure ??. R2OMC produces samples that closely match the ground truth posterior, with a slightly greater spread again due to the ϵ approximation, while ROMC and NPE continue to disperse their samples along the 45° line.

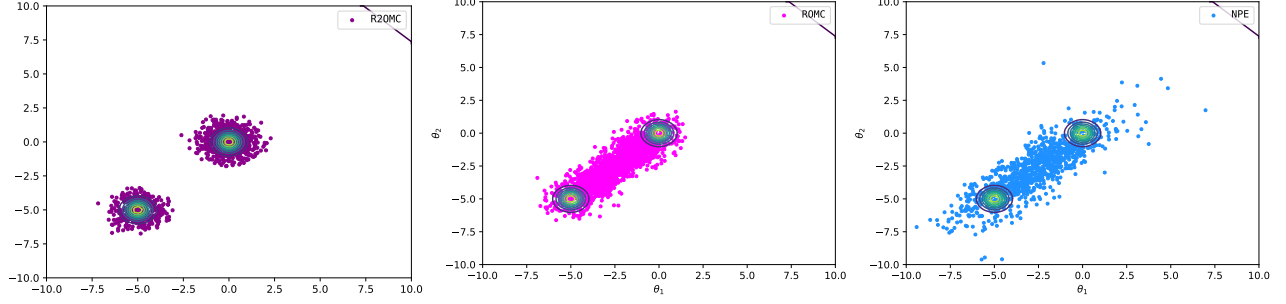


Figure 5: Caption

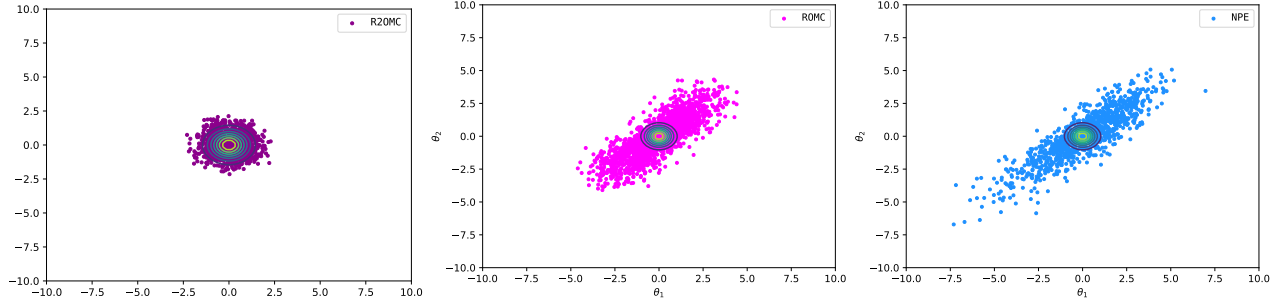


Figure 6: Caption

3.1.2 High-dimensional dataset

At the high-dimensional dataset, we vary the dimensionality of the parameter space in $D_{\theta} \in \{2, 5, 10, 20, 25, 30\}$. In each case, we set the observation to $\mathbf{y}^o = (2^2, \dots, 2^2)$. When adding distractors, we append 100 zeros in the end of the observation, i.e., $\mathbf{y}^o = (2^2, \dots, 2^2, 0, \dots, 0)$. The distractors do not affect the ground truth posterior, which remains the same as in the standard setup. Below we provide the ground truth posterior for each setup.

S_{linear} : The simulator is $S_{\text{linear}} : \mathbf{y} \sim \mathcal{N}(\boldsymbol{\theta}, 1.5^2 \odot \mathbf{I})$. The ground truth posterior is:

$$p(\boldsymbol{\theta}|\mathbf{y}^o) \propto p(\boldsymbol{\theta})p(\mathbf{y}^o|\boldsymbol{\theta}) \quad (23)$$

$$\propto \mathcal{U}(-15, 15)\mathcal{N}(\mathbf{y}^o; \boldsymbol{\theta}, 1.5^2 \odot \mathbf{I}) \quad (24)$$

$$\propto \mathcal{N}(\mathbf{y}^o; \boldsymbol{\theta}, 1.5^2 \odot \mathbf{I}) \quad (25)$$

$$\propto \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}^o, 1.5^2 \odot \mathbf{I}) \quad (26)$$

$$\propto \mathcal{N}(\boldsymbol{\theta}; \mathbf{2}, 1.5^2 \odot \mathbf{I}) \quad (27)$$

$S_{\text{MoG}(\Sigma_1=\Sigma_2)}$: The simulator is $S_{\text{MoG}(\Sigma_1=\Sigma_2)} : \mathbf{y} \sim \frac{1}{2} (\mathcal{N}(\boldsymbol{\theta}, 0.1^2 \mathbf{I}) + \mathcal{N}(-\boldsymbol{\theta}, 0.1^2 \mathbf{I}))$. The ground truth posterior is:

$$p(\boldsymbol{\theta}, \mathbf{y}^o) \propto p(\boldsymbol{\theta})p(\mathbf{y}^o|\boldsymbol{\theta}) \quad (28)$$

$$\propto \mathcal{U}(-15, 15) (\mathcal{N}(\mathbf{y}^o; \boldsymbol{\theta}, 0.1^2 \mathbf{I}) + \mathcal{N}(\mathbf{y}^o; -\boldsymbol{\theta}, 0.1^2 \mathbf{I})) \quad (29)$$

$$\propto (\mathcal{N}(\mathbf{y}^o; \boldsymbol{\theta}, 0.1^2 \mathbf{I}) + \mathcal{N}(\mathbf{y}^o; -\boldsymbol{\theta}, 0.1^2 \mathbf{I})) \quad (30)$$

$$\propto (\mathcal{N}(\boldsymbol{\theta}; \mathbf{y}^o, 0.1^2 \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{y}^o, 0.1^2 \mathbf{I})) \quad (31)$$

$$\propto (\mathcal{N}(\boldsymbol{\theta}; \mathbf{2}, 0.1^2 \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{2}, 0.1^2 \mathbf{I})) \quad (32)$$

which is a mixture of two Gaussians with centers at $\mu_1 = (2, \dots, 2)$ and $\mu_2 = (-2, \dots, -2)$ and covariances $\Sigma_1 = \Sigma_2 = 0.1^2 \mathbf{I}$.

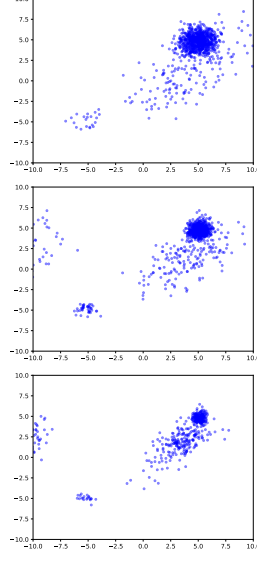


Figure 7: Caption

$S_{\text{MoG}(\Sigma_1 \neq \Sigma_2)}$. The simulator is $S_{\text{MoG}(\Sigma_1 \neq \Sigma_2)} : \mathbf{y} \sim \frac{1}{2} (\mathcal{N}(\boldsymbol{\theta}, 0.1^2 \odot \mathbf{I}) + \mathcal{N}(-\boldsymbol{\theta}, \mathbf{I}))$. The ground truth posterior is:

$$p(\boldsymbol{\theta}, \mathbf{y}^o) \propto p(\boldsymbol{\theta})p(\mathbf{y}^o|\boldsymbol{\theta}) \quad (33)$$

$$\propto \mathcal{U}(-\mathbf{15}, \mathbf{15}) (\mathcal{N}(\mathbf{y}^o; \boldsymbol{\theta}, 0.1^2 \mathbf{I}) + \mathcal{N}(\mathbf{y}^o; -\boldsymbol{\theta}, \mathbf{I})) \quad (34)$$

$$\propto \mathcal{N}(\mathbf{y}^o; \boldsymbol{\theta}, 0.1^2 \mathbf{I}) + \mathcal{N}(\mathbf{y}^o; -\boldsymbol{\theta}, \mathbf{I}) \quad (35)$$

$$\propto \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}^o, 0.1^2 \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{y}^o, \mathbf{I}) \quad (36)$$

$$\propto \mathcal{N}(\boldsymbol{\theta}; \mathbf{2}, 0.1^2 \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{2}, \mathbf{I}) \quad (37)$$

which is a mixture of two Gaussians with centers at $\mu_1 = (2, \dots, 2)$ and $\mu_2 = (-2, \dots, -2)$ and covariances $\Sigma_1 = 0.1^2 \mathbf{I}$ and $\Sigma_2 = \mathbf{I}$.

Results

3.1.3 Multi-modal posterior

In this setup, we have a simulator $S_{\text{cartesian}} : \mathbf{y} \sim \frac{1}{|\mathcal{M}|} \sum_{\mu_i \in \mathcal{M}} \mathcal{N}(\mu_i, 0.8^2 \odot \mathbf{I})$, where $\mathcal{M} = \{\{\pm\theta_1\} \times \dots \times \{\pm\theta_{D_\theta}\}\}$. The observation is $\mathbf{y}^o = \mathbf{2}$. The ground truth posterior is:

$$p(\boldsymbol{\theta}|\mathbf{y}^o) \propto p(\boldsymbol{\theta})p(\mathbf{y}^o|\boldsymbol{\theta}) \quad (38)$$

$$\propto \mathcal{U}(-\mathbf{15}, \mathbf{15}) \frac{1}{|\mathcal{M}|} \sum_{\mu_i \in \mathcal{M}} \mathcal{N}(\mathbf{y}^o; \mu_i, 0.8^2 \odot \mathbf{I}) \quad (39)$$

$$\propto \sum_{\mu_i \in \mathcal{M}} \mathcal{N}(\mathbf{y}^o; \mu_i, 0.8^2 \odot \mathbf{I}) \quad (40)$$

$$\propto \sum_{\mu_i \in \mathcal{M}} \mathcal{N}(\mu_i; \mathbf{y}^o, 0.8^2 \odot \mathbf{I}) \quad (41)$$

$$\propto \sum_{c \in \mathcal{C}} \mathcal{N}(\boldsymbol{\theta}; c, 0.8^2 \odot \mathbf{I}) \quad (42)$$

where $\mathcal{C} = \underbrace{\{\{\pm 2\} \times \dots \times \{\pm 2\}\}}_{D_{\theta} \text{ times}}$. The ground truth posterior is a mixture of $2^{D_{\theta}}$ Gaussians with centers at $(\pm 2, \dots, \pm 2)$ and covariances $0.8^2 \mathbf{I}$.

4 Additional results

5 FORMATTING INSTRUCTIONS

To prepare a supplementary pdf file, we ask the authors to use `aistats2025.sty` as a style file and to follow the same formatting instructions as in the main paper. The only difference is that the supplementary material must be in a *single-column* format. You can use `supplement.tex` in our starter pack as a starting point, or append the supplementary content to the main paper and split the final PDF into two separate files.

Note that reviewers are under no obligation to examine your supplementary material.

6 MISSING PROOFS

The supplementary materials may contain detailed proofs of the results that are missing in the main paper.

6.1 Proof of Lemma 3

In this section, we present the detailed proof of Lemma 3 and then [...]

7 ADDITIONAL EXPERIMENTS

If you have additional experimental results, you may include them in the supplementary materials. p

7.1 The Effect of Regularization Parameter

Our algorithm depends on the regularization parameter λ . Figure 1 below illustrates the effect of this parameter on the performance of our algorithm. As we can see, [...]